

Properties of Convex Functions

$$x^* = -2/3$$

$$f(x) = |x|$$



Looking at the examples:

$$f(x) = 3x^2 + 4x - 2$$

$$f(x_1, x_2) = 10|x_1| + |x_2|$$

We see three key differences between gradient descent and the subgradient method.

$$x_{t+1} = x_t - \eta(6x_t + 4)$$

gradient descent improves at every iteration, unlike sub-gradient ✓

gradient descent can take a "big" step size: self-tuning property ✓

gradient descent takes bigger steps when far away ✓

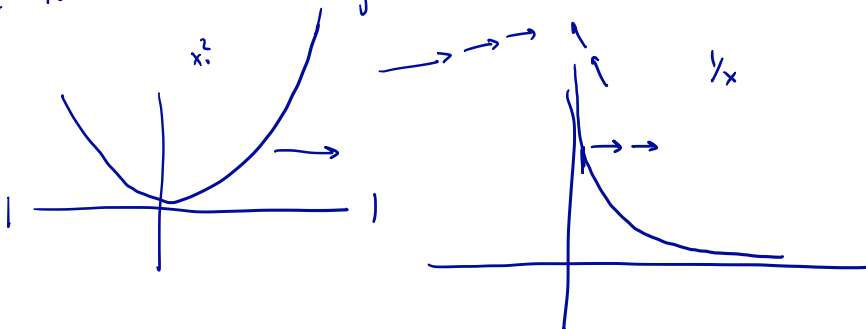
We formalize these through properties some convex functions have. We will then use these properties directly to show different convergence rates, depending on which properties a particular convex function enjoys.

Basic assumptions we make throughout

1. We don't start "so 'ly" far away from x^* .

We will see: $\|x_1 - x^*\|_2$

2. Functions are not "so 'ly" steep.



Smoothness (1/2)

definition and quadratic upper bound

"self-tuning" : $\|\nabla f(x)\| \rightarrow 0$ as $x \rightarrow x^*$

$$\text{Need: } \|\nabla f(x) - \nabla f(y)\|_2 \leq \beta \cdot \|x - y\|_2$$

Def'n: A function f is called β -smooth if its gradient is Lipschitz continuous with parameter β : $\|\nabla f(x) - \nabla f(y)\|_2 \leq \beta \cdot \|x - y\|_2$ (*)

Claim : If $f(x)$ is β -smooth, then : $\underbrace{\frac{\beta}{2} \|x\|_2^2 - f(x)}_{g(x)}$ is convex.

Recall that monotone $\Rightarrow$ convex.

$$\begin{aligned} \langle \nabla g(x) - \nabla g(y), x - y \rangle &= \langle \beta(x - y) - (\nabla f(x) - \nabla f(y)), x - y \rangle \\ &= \beta \|x - y\|_2^2 - \underbrace{\langle \nabla f(x) - \nabla f(y), x - y \rangle}_{\|\nabla f(x) - \nabla f(y)\|_2 \cdot \|x - y\|_2} \\ &\geq \beta \|x - y\|_2^2 - \|\nabla f(x) - \nabla f(y)\|_2 \cdot \|x - y\|_2 \\ &\geq \beta \cdot \|x - y\|_2^2 - \beta \cdot \|x - y\|_2 \cdot \|x - y\|_2 = 0 \end{aligned}$$

$$\langle u, v \rangle \leq \|u\|_2 \cdot \|v\|_2$$

Smoothness (2/2)

definition and quadratic upper bound

We just showed: $g(x) = \frac{\beta}{2} \|x\|_2^2 - f(x)$ is convex

$$g(y) \geq g(x) + \langle \nabla g(x), y - x \rangle$$

$$\frac{\beta}{2} \|y\|_2^2 - f(y) \geq \frac{\beta}{2} \|x\|_2^2 - f(x) + \langle \beta x - \nabla f(x), y - x \rangle$$

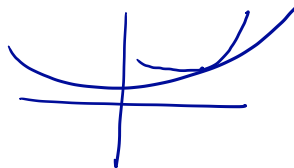
$$\begin{aligned} f(y) &\leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y\|_2^2 - \beta \langle x, y \rangle - \frac{\beta}{2} \|x\|_2^2 + \beta \|x\|_2^2 \\ &= f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2 \end{aligned}$$

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\beta}{2} \|y - x\|_2^2$$

twice differentiable: upper bound on Hessian

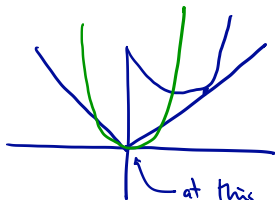
If f is twice diff'ble: f convex $\Leftrightarrow \nabla^2 f(x) \succeq 0$

If f is β -smooth: $\nabla^2 f(x) \preceq \beta \cdot I$



Smoothness: two examples

Absolute value



at this point $x=0$, no quadratic
upper bounds the fn.

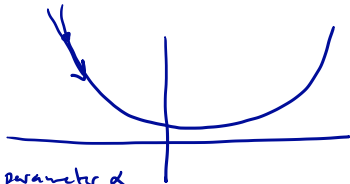
Quadratic (eigenvalues)

$$f(x) = x^T Q x + \underbrace{g^T x + c}$$

Exercise: $f(x)$ is β -smooth for any $\beta \geq \underbrace{2 \lambda_{\max}(Q)}$

Strong Convexity

definition and quadratic lower bound



Def 1: A function f is strongly convex w/ parameter α if $g(x) = f(x) - \frac{\alpha}{2} \|x\|_2^2$ is convex.

$$g(y) \geq g(x) + \langle \nabla g(x), y - x \rangle$$

$$\Rightarrow f(y) - \frac{\alpha}{2} \|y\|_2^2 \geq f(x) - \frac{\alpha}{2} \|x\|_2^2 + \langle \nabla f(x) - \alpha x, y - x \rangle$$

$$\Rightarrow f(y) \geq \underbrace{f(x) + \langle \nabla f(x), y - x \rangle}_{\text{standard convexity}} + \underbrace{\frac{\alpha}{2} \|y - x\|_2^2}_{\text{strong convexity}}$$

twice differentiable: lower bound on Hessian

$$\begin{aligned} f(x) \text{ is } \alpha\text{-strongly convex} &\Leftrightarrow \nabla^2 f(x) \succeq \alpha I \\ &\Leftrightarrow (\nabla^2 f(x) - \alpha I) \succeq 0 \end{aligned}$$

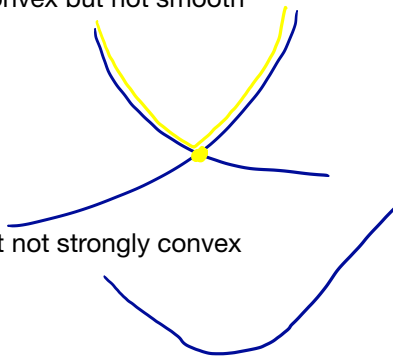
Examples

Quadratic

$$f(x) = x^T Q x$$

$f(x)$ is α -st. convex
 $\alpha = 2\lambda_{\min}(Q)$.

Strongly convex but not smooth



Smooth but not strongly convex